

Speaker 3 of 5 — Data Cleaning

Career Compass SG · Module 1 Assignment · Group 7

Your time: up to 3 minutes · You have the strongest story in the presentation

Your job in one sentence

Prove we understood this data instead of just loading it. Data handling carries the biggest weight in the marking (25%), and you own the best story in the whole talk — the discovery that the dataset was secretly two files stitched together.

Where you sit in the flow

	Speaker	Topic
	1	Dashboard overview
	2	Project technology stack
→ YOU	3	Data cleaning
	4	Data analysis 1 — Demand vs Competition
	5	Data analysis 2 — Career Recommender

Speaker 2 hands you: "...this dataset had a trap hidden inside it." **Your handover to Speaker 4:** "So that's the data we can trust. Now let's see what it tells us about demand and competition."

Time rule: if you are running late, cut Part 2. **Part 3 is the story that must be told** — protect it.

Script

Part 1 — Our cleaning principle (0:00 – 0:50)

(Slide: cleaning examples — show 3, not all 11)

We wrote eleven cleaning rules. The one principle behind all of them:

If a value is broken, blank the value — but keep the row.

A posting with a broken salary still proves **a job existed** — deleting the whole row would break our job counts just to make salaries look clean. Three examples:

- One column was **completely empty** in all 1.05 million rows — dropped.
- About **10,000 postings claimed salaries like \$180,000 a month** — yearly figures in a monthly field. We blanked the salary, kept the posting.
- The most extreme 1% of salaries were **capped, not deleted**.

After all the cleaning, **99.6% of the rows survive**.

Part 2 — Two traps that nearly caught us (0:50 – 1:30)

(Slide: the two traps)

Two traps worth sharing, because they could catch anyone in this room.

Trap one: the duplicate check lied. pandas reported 289 duplicate job IDs — all 289 were **empty cells compared with each other**. Real duplicates: **zero**.

Trap two: one job can belong to up to three categories — 1.69 on average. Count rows after splitting by category and you double-count. We made this mistake ourselves: for an hour our front page said **1.66 million postings instead of 1.04 million** — and it looked completely believable.

Part 3 — The discovery that changed everything (1:30 – 2:50)

(Slide: the two-line engagement chart — speak slowly here)

Now the big one.

As a routine check, we plotted views and applications **over time**. We were not looking for a problem. Look at what happens at **June 2023**.

Average views per posting **fall from about 111 to about 5**. The share of postings with **zero applications jumps from 18% to 79%**.

Did Singaporeans stop applying for jobs? Of course not. **This dataset is two separate downloads stitched together**. Postings after June 2023 were captured the moment they were posted — their counters **never had time to grow**. They're frozen at zero.

So our design decision: **job counts and salaries use all 1.04 million rows, but competition uses only postings up to June 2023** — about 200,000 — and every page says so on screen.

If we had missed this, every track would have looked like it had **no competition at all** — and our recommender would have sent people into the most crowded markets in Singapore.

That is the difference between a dashboard and a wrong dashboard.

So that's the data we can trust. Now — what it tells us about demand and competition.

Numbers you must know cold

Figure	Value
Rows read → kept	1,048,585 → 1,044,597 (99.6%)
Impossible salaries blanked	~10,000
Salary rows capped (top/bottom 1%)	19,647
Real duplicate job IDs	0 (the "289" were blanks)
Categories per job	1.69 average, max 3
The break	views 111 → 5, zero-application 18% → 79%, at June 2023
Competition window	postings up to 30 June 2023 (~203,702)

Questions you own

"Why cap outliers instead of deleting them?"

Deleting the row also deletes the job from our demand counts. Capping keeps the counts honest while stopping extreme values from distorting averages.

"How do you know the June 2023 break is a data problem and not real?"

Because only the engagement counters break — posting volume, salaries, and categories all continue smoothly across that date. A real market collapse would show up everywhere.

"Isn't using only 200,000 postings for competition too few?"

It's still two hundred thousand postings — plenty. And the alternative was using numbers we *know* are frozen at zero. Better a smaller honest number than a bigger wrong one.

"Why drop rows with missing values instead of filling them in?"

We checked first: the missing values were all on the *same* rows — completely empty lines with no ID, no title, no dates. There was nothing left to fill in from.

Rehearsal checklist

- [] Timed at **3:00 or under** — and you know Part 2 is the part to cut if late
- [] You can tell the June 2023 story from memory, without the slide
- [] You can explain in one sentence why the counters are frozen, not wrong
- [] You are comfortable admitting our own 1.66-million bug — owning it reads as competence
- [] Handover line practised